IP CORES For FPGA Designs

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Intellectual property (IP) cores are standalone modules that can be used in any field programmable gate array (FPGA). These are developed using HDL languages like VHDL, Verilog and System Verilog, or HLS like C.

IP cores are part of the growing electronic design automation (EDA) industry. In this article, these will be discussed with respect to SRAM based FPGAs.

Let us take an example of universal asynchronous receiver-transmitter (UART) IP block, which is intended to be used in different applicatons. The developed UART IP core module should:

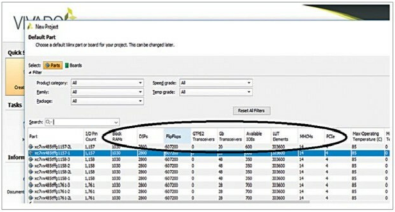
- Meet basic UART functionalities
- Be portable, so that it can be used in any vendor technologies; for example, Xilinx/Altera as plug and play
- Have user-configurable parameters (like baud rate)
- Have processor interface/generic parameter file to modify configurations as required
- Provide IP data sheet

Types of IP cores

IP cores can be categorised as hard IP core, firm IP (semi-hard IP) core and soft IP core.

Hard IP cores. These are part of the FPGA-independent modules; for example, PCIe or Ethernet IP modules available in Xilinx FPGA. You have to configure the location and provide interface connectivity with other modules, clocks and resets.

Fig. 1: Xilinx Vivado - FPGA selection overview includes hard blocks



Since these blocks are already part of the FPGA device, these will not be taken into account while calculating the utilisation of the slice logic report. In the utilisation summary, these will be counted as the number of PCIe/Ethernet blocks used. Because of a fixed location in the FPGA, these cores cannot be ported to other FPGAs. Neither can these be reused like HDL components, if already used in the FPGA.

Fig. 1 shows Xilinx Vivado tool - FPGA selection window for Virtex-7 FPGA with internal hard IP details for creating the project. The number of hard IPs may vary between different FPGA families.

Fig 2. shows the dedicated location of the hard IP in Virtex-7 FPGA.

Unmarked (not labelled) areas (between DSP slice and block RAM, or block RAM and PCIe, or PCIe and transceivers) in the FPGA (Fig. 2) contain large distribution of flip-flops, latches, multiplexers, LUTs, etc. Soft IP cores or custom logic are implemented in these areas.

Advantages include:

- Timing violations minimised
- No extra cost, that is, cost of the hard block is included in the FPGA; hence, can be considered as low-cost compared to other two types of IP cores
- No individual licence, except for compiler tool licence
- RTL code maintenance reduced
- No extra documentation required; for each IP level, documentation provided by vendor
- For slice/LUT summary, hard IP cores not considered
- Low dynamic and static power can be achieved if hard IP blocks are used in the designs
- Functionality and performance guaranteed
- Fully-tested and with known errors/limitations, if any, as per documents provided by vendor

Disadvantages include:

- Not portable; these are highly optimised and targeted at specific FPGAs